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Data Analyst Intern

Day 11

Tasks

1. AI Recommendation System Research

AI Recommendations

AI Recommendation Topic	Explanation	Example for Raritone
Personalized Recommendations	AI shows products based on user interests and behavior	Suggest dresses based on browsing history
Similar Product Recommendations	AI suggests products similar to viewed or tried items	Recommend similar kurtis after virtual try-on
User Preference Tracking	AI tracks categories, colors, brands, and styles users prefer	Track users who often view western wear
AI Shopping Behavior Analysis	AI studies clicks, wishlist, try-on, and purchases	Identify users interested in party wear
Trending Recommendation Logic	AI promotes trending and high-engagement products	Show trending outfits on homepage
Wishlist-Based Recommendations	AI recommends products based on wishlist saves	Suggest matching accessories
Purchase Prediction	AI predicts users likely to purchase soon	Offer discount to high-intent users
Style Matching Recommendations	AI suggests complete outfit combinations	Match tops with jeans or footwear

Recommendation systems work by analyzing user behavior such as product views, clicks, wishlist activity, purchases, session duration, and virtual try-on usage. AI identifies patterns in user interests and suggests products that match their preferences. Fashion apps also use similar product recommendations, trending product analysis, and personalized feeds to improve product discovery and engagement.

5 Ways AI Can Improve Raritone Shopping Experience

No.	AI Improvement	Benefit for Raritone
1	Personalized product recommendations	Users see products matching their style and interests
2	Similar outfit suggestions after try-on	Increases product discovery and add-to-cart rate
3	AI-based trend analysis	Helps promote trending fashion categories
4	Smart notifications based on user behavior	Improves retention and re-engagement
5	Purchase prediction models	Helps target users likely to purchase soon

2. User Behavior Prediction Analysis

User ID	Engagement Level	Session Duration	Product Interactions	Virtual Try-On Usage	Purchase Probability	Retention Risk	Preferred Category
U101	High	8.2	45	12	Very High	Low	Western Wear
U102	Medium	5.1	20	4	Medium	Medium	Casual Wear
U103	Low	2.3	8	1	Low	High	Accessories
U104	High	9	50	15	Very High	Low	Ethnic Wear
U105	Medium	4.8	18	3	Medium	Medium	Streetwear
U106	Low	1.9	5	0	Very Low	Very High	Casual Wear
U107	High	7.5	38	10	High	Low	Party Wear

Prediction Analysis

Prediction Insight	Observation	Suggested Action
High engagement users are more likely to purchase	Users with longer sessions and more interactions show strong purchase intent	Show personalized offers and recommendations
Users with low try-on usage may lose interest	Users who do not use virtual try-on engage less	Promote try-on tutorials and reminders
Low engagement users have higher retention risk	Users with short sessions and low interactions may stop using the app	Send reactivation campaigns and personalized notifications
Users prefer categories based on browsing behavior	Different users show interest in different fashion styles	Customize homepage recommendations

User behavior prediction helps Raritone identify which users are likely to purchase, which users may stop using the app, and which products users may prefer. By analyzing engagement level, session duration, product interactions, and virtual try-on usage, Raritone can improve personalization, retention, and conversion.

3. Virtual Try-On Analytics Task

Virtual Try-On Analytics

Virtual Try-On Topic	Observation	Importance for Raritone
Why Users Use Virtual Try-On	Users want to visualize clothes before buying	Improves purchase confidence
Most Engaged Features	Outfit preview, color variation, size visualization	Helps users compare styles and fit
Purchase Confidence Impact	Users feel more confident after trying clothes virtually	Can increase add-to-cart and purchases
Return Reduction Potential	Virtual try-on may reduce size confusion	Can reduce product returns
User Experience Improvement	Interactive shopping increases engagement	Makes shopping more enjoyable

Benefits of Virtual Try-On

Benefit	Explanation
Better shopping confidence	Users can visualize clothing before purchase
Higher engagement	Users spend more time interacting with products
Reduced size confusion	Users understand fit and appearance better
Increased product discovery	Users may try multiple outfits
Improved conversion rate	Confident users are more likely to purchase

Suggested Improvements for Raritone

No.	Suggested Improvement	Why It Helps
1	Improve try-on speed and accuracy	Better experience increases user satisfaction
2	Add style recommendations after try-on	Encourages users to explore more products
3	Add social sharing for try-on looks	Increases engagement and brand awareness
4	Improve mobile try-on experience	Most users shop through mobile devices
5	Add AI styling suggestions	Makes the shopping experience more personalized

4. Product Performance Analysis

Product Performance

Product Name	Category	Product Views	Try-On Usage	Wishlist Saves	Add to Cart	Purchases	Engagement Level
Floral Kurti	Ethnic Wear	4200	950	620	410	180	High
Denim Jacket	Western Wear	3800	870	540	360	150	High
Party Dress	Party Wear	4500	1100	700	460	210	Very High
Casual T-Shirt	Casual Wear	2600	320	210	180	70	Medium
Sneakers	Footwear	3100	280	190	150	60	Medium
Saree Collection	Ethnic Wear	4000	920	610	390	170	High
Office Blazer	Workwear	1900	120	80	60	20	Low

Product Performance Analysis

Insight	Observation	Suggested Action
Party Wear has highest engagement	Users spend more time and use try-on more	Promote party wear collections and recommendations
Office Blazer has low engagement	Fewer views and try-on interactions	Improve product images and targeting
Ethnic Wear performs strongly	Users engage heavily with kurtis and sarees	Expand ethnic wear collections
Products with high try-on usage perform better	Users who use try-on are more likely to purchase	Encourage try-on before purchase

5. Smart Shopping Insights Dashboard

Smart Insights Dashboard

Raritone Day 11 Smart Shopping Insights Dashboard				
Most Preferred Category	Highest Engagement Product	Avg Try-On Usage	High Purchase Probability Users	Key Retention Risk
Ethnic Wear	Party Dress	651	3 Users	Low Engagement Users

6. Data Visualization Task

Chart 1 — Product Preference Graph

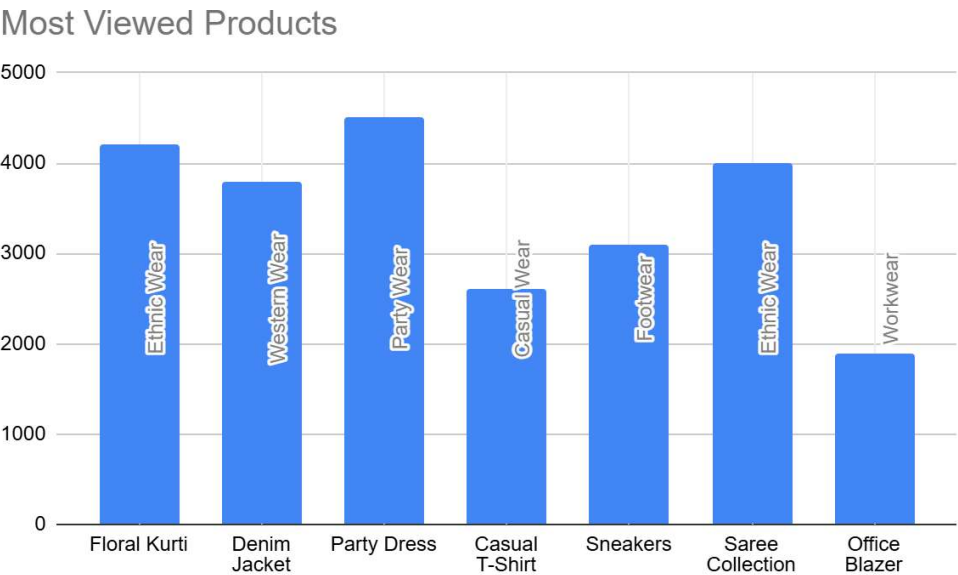


Chart 2 — Engagement vs Purchase Graph

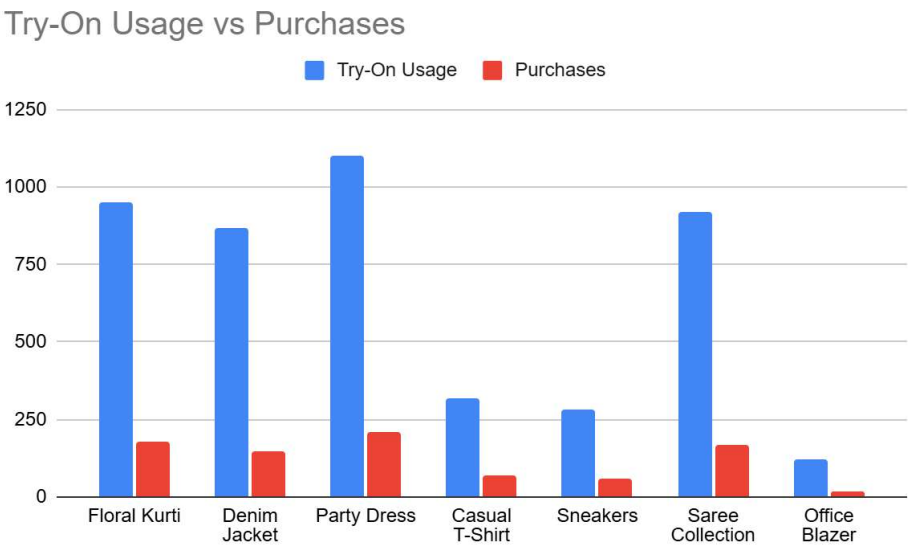
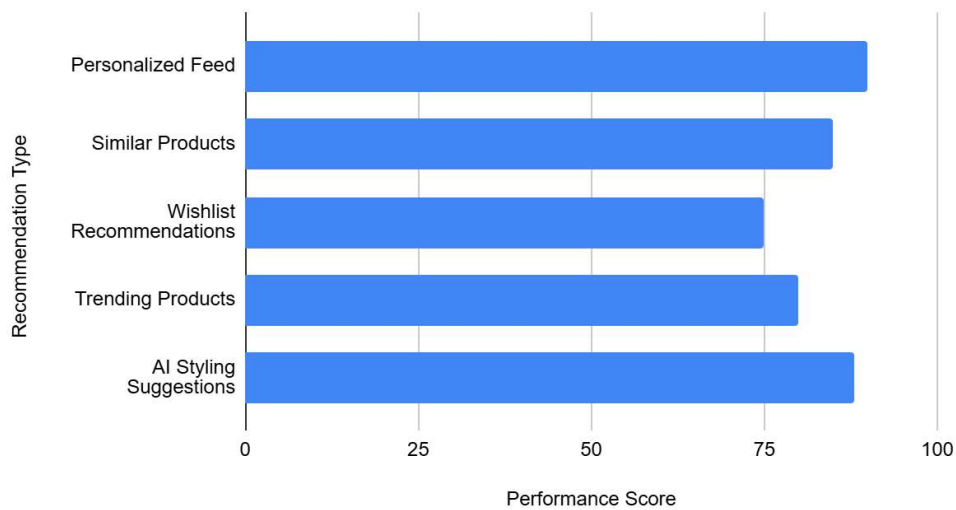


Chart 3 — Recommendation Analytics Graph

Recommendation Type	Performance Score
Personalized Feed	90
Similar Products	85
Wishlist Recommendation	75
Trending Products	80
AI Styling Suggestions	88

Recommendation Performance



7. Daily Summary

Today, I completed the Day 11 Data Analyst Intern task for Raritone. I researched AI recommendation systems used in fashion apps, including personalized recommendations, similar product suggestions, user preference tracking, and AI-based shopping behavior analysis.

I created a user behavior prediction report using engagement level, session duration, product interactions, and virtual try-on usage to identify users likely to purchase or stop using the app.

I also analyzed virtual try-on usage, benefits, and engagement impact, and suggested improvements for Raritone's try-on experience. In addition, I created a product performance analysis sheet showing the most viewed, most tried-on, and highest engagement products and categories.

Finally, I updated the Smart Shopping Insights Dashboard with recommendation analytics, prediction insights, product preference charts, virtual try-on engagement metrics, and recommendation performance graphs.

Overall, Day 11 helped me understand how AI recommendations, user behavior prediction, and virtual try-on analytics can improve shopping experience, engagement, and retention for Raritone.